

Measurement of Single Qubit State (Lecture 2)

Course: Survey of Quantum Technologies & Applications (PH 40001)

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Measurement

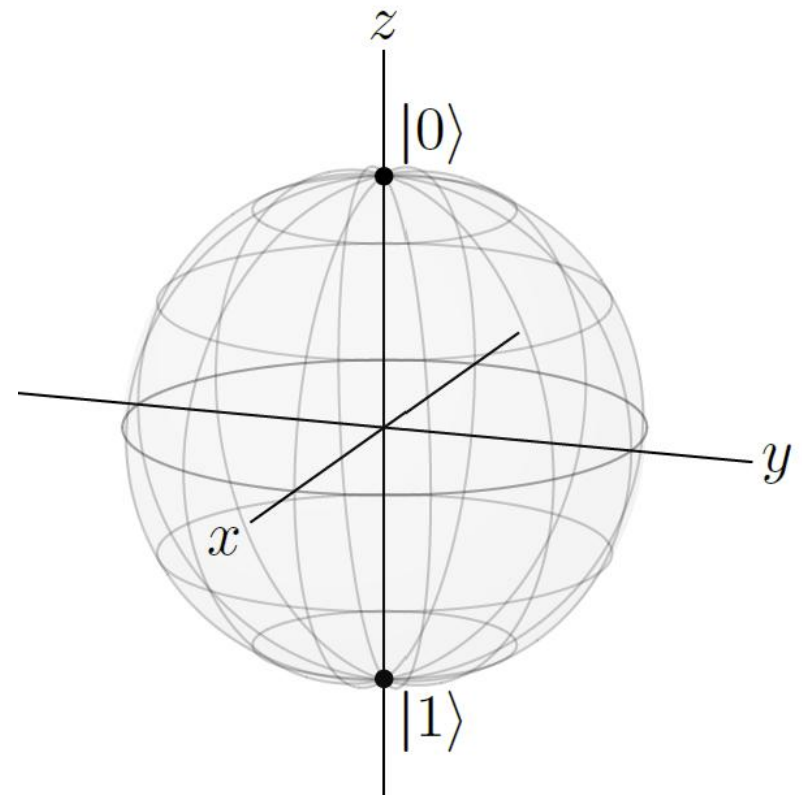
- Measurement turns amplitudes into classical probabilities.
- When we measure a qubit state, we get $|0\rangle$ and $|1\rangle$, each with some probability.

Measurement Basis

- Measurement in Z-basis
 - Z-basis: $\{|0\rangle, |1\rangle\}$ lies on the z-axis
- Measurement in X-basis
 - X-basis: $\{|+\rangle, |-\rangle\}$ lies on the x-axis
- Measurement in Y-basis
 - Y-basis: $\{|i\rangle, |-i\rangle\}$ lies on the y-axis

Measurement in Z-basis

A measurement returns classical outcome: $|0\rangle$ or $|1\rangle$.



Example 1

If we measure the qubit state $|\Psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, what are the probabilities of getting states $|0\rangle$ and $|1\rangle$?

The probability of getting $|0\rangle$ is

$$|\alpha|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

The probability of getting $|1\rangle$ is

$$|\beta|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Example 2

If we measure the qubit state $|\Psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/6}|1\rangle)$, what are the probabilities of getting states $|0\rangle$ and $|1\rangle$?

The probability of getting $|0\rangle$ is

$$|\alpha|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

The probability of getting $|1\rangle$ is

$$|\beta|^2 = \left| \frac{e^{i\pi/6}}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Example 3

If we measure the qubit state $|\Psi\rangle = \frac{2}{3}|0\rangle + \frac{1-2i}{3}|1\rangle$, what are the probabilities of getting states $|0\rangle$ and $|1\rangle$?

The probability of getting $|0\rangle$ is

$$|\alpha|^2 = \left|\frac{2}{3}\right|^2 = \frac{4}{9}$$

The probability of getting $|1\rangle$ is

$$|\beta|^2 = 1 - |\alpha|^2 = \frac{5}{9}$$

Example 4

If we measure the qubit state $|\Psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$ in the X-basis, what states can we get and what would be their probabilities ?

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle), \quad |1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle).$$

Substituting into the state of our qubit,

$$\begin{aligned} \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle &= \frac{\sqrt{3}}{2} \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \frac{1}{2} \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}}|+\rangle + \frac{\sqrt{3}-1}{2\sqrt{2}}|-\rangle. \end{aligned}$$

Example 4

The probability of measuring $|+\rangle$ is

$$\left| \frac{\sqrt{3} + 1}{2\sqrt{2}} \right|^2 = \frac{\sqrt{3} + 2}{4} \approx 0.93$$

the probability of measuring $|-\rangle$ is

$$\left| \frac{\sqrt{3} - 1}{2\sqrt{2}} \right|^2 = \frac{-\sqrt{3} + 2}{4} \approx 0.07$$

Repeated Measurements in Quantum Circuits

Main idea

In quantum computing, “repeated measurement” can mean two different things.

- 1) Measuring the same physical qubit again after it has already been measured.
- 2) Repeating the entire experiment many times.
These repetitions are called shots.

Repeated Measurements in Quantum Circuits

Important Concept

Probabilities such as $P(0)$ and $P(1)$ are usually interpreted from many identical repetitions of the experiment.

For the same qubit measured twice in the same basis, the second result is determined by the collapse from the first measurement.

Measurement collapses the qubit.

Case 1: Consecutive Measurement of the Same Qubit

Initial state
 $|\Psi\rangle = (2/3)|0\rangle + ((1 - 2i)/3)|1\rangle$

1st measurement
Outcome obtained:
 $|0\rangle$

State after collapse
 $|\Psi_{\text{after}}\rangle = |0\rangle$

2nd Z-basis
measurement
 $P(0)=1, P(1)=0$

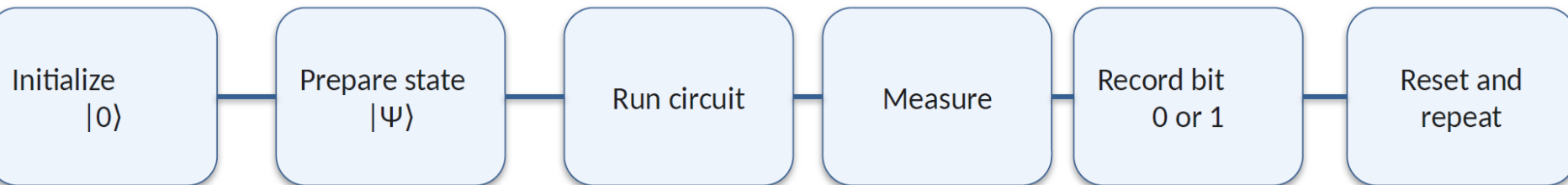
Conclusion

If you measure the same qubit again immediately in the same basis, you get the same result with probability 1.

Case 2: Shots in a Quantum Circuit

Definition of a shot

One shot = initialize qubit(s) $\rightarrow$ prepare the state $\rightarrow$ run the circuit $\rightarrow$ measure $\rightarrow$ record the outcome.



Important point

A new shot does not reuse the already-collapsed qubit. Instead, the experiment starts again from the beginning with a fresh preparation of the state.

Why Probabilities Usually Mean “Many Shots”

Theoretical probabilities

For $|\Psi\rangle = (2/3)|0\rangle + ((1 - 2i)/3)|1\rangle$:

$$P(0) = 4/9 \approx 0.444$$

$$P(1) = 5/9 \approx 0.556$$

Interpretation using shots

If we run 1000 shots, we expect approximately:

$$\text{Count}(0) \approx 444$$

$$\text{Count}(1) \approx 556$$

Actual counts vary slightly because each shot is random.

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